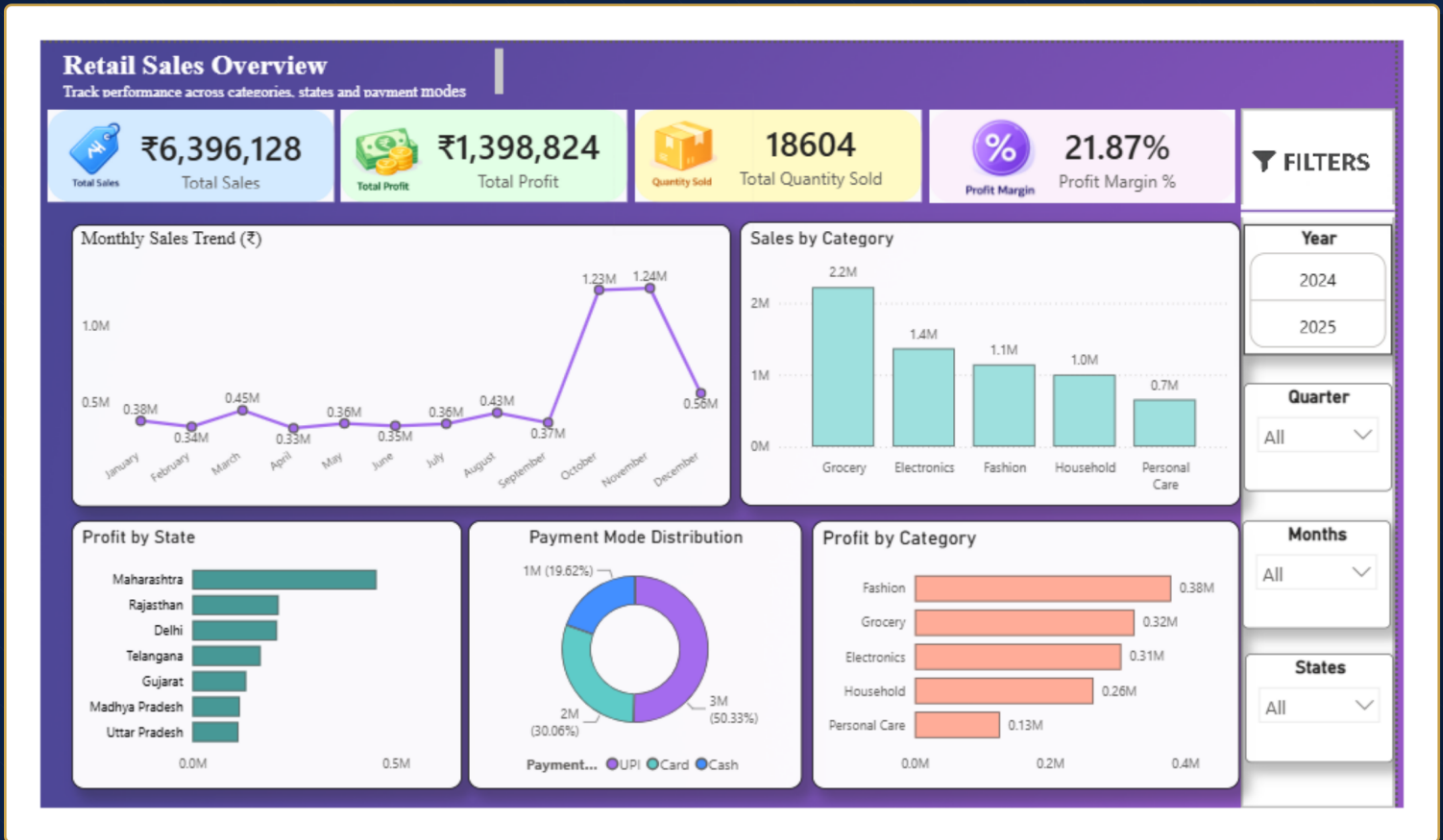


RETAIL SALES ANALYSIS

Business Intelligence Report using Excel, SQL Server & Power BI

EXCEL • SQL SERVER • POWER BI



Retail Sales Overview — Power BI Dashboard

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PROJECT DURATION

June – July 2026

TOOLS USED

Excel • SQL • Power BI

D-MART RETAIL TRANSACTIONS DATASET • 5,000 ORDERS • FY 2024–2025

Executive Summary

For this project, I analysed 5,000 D-Mart retail transactions from FY 2024–2025 using Excel, SQL, and Power BI, with the goal of understanding where the business actually makes its money — by category, product, region, customer type, payment method, and time period — and turning those patterns into recommendations a retail team could act on.

The headline numbers are healthy: ₹63.96 lakh in total sales and ₹13.99 lakh in profit, a margin of 21.87%. But the more interesting story is underneath that. Grocery sells the most, yet Fashion is the real profit driver, earning more than double Grocery's margin. Maharashtra, and Mumbai in particular, carries roughly a third of the business on its own. Regular customers and UPI payments each make up close to two-thirds of transaction value, which says a lot about where retention and payment-experience efforts should go. October and November alone account for nearly 40% of yearly sales, pointing to a clear festive-season pattern. I also compared 2024 against 2025 and found a modest, consistent softening across sales, profit, orders, and quantity — not dramatic, but worth watching.

These findings led to seven practical recommendations, covering margin management, seasonal planning, regional investment, payment experience, customer retention, product pricing, and early trend monitoring (Section 8). I cross-checked every major figure across Excel, SQL, and Power BI before including it here, so the numbers hold up no matter which tool you check them against.

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1. Project Overview

Business Background

D-Mart is a large-format Indian retail chain operating across major cities, selling everyday products spanning grocery, personal care, household, fashion, and electronics. This project analyses a transaction-level extract from D-Mart's retail operations to understand sales performance, profitability, and customer/regional behaviour.

Problem Statement

Retail leadership needs a clear, data-driven view of where revenue and profit are being generated — by category, product, region, customer type, payment method, and time period — in order to guide inventory, marketing, and regional investment decisions.

Objectives

- Consolidate 5,000 retail transactions into a clean, analysis-ready dataset.
- Use Excel (Pivot Tables and a Dashboard) to summarise sales, profit, and quantity trends.
- Use SQL to answer 40 specific business questions across sales, product, category, region, customer, payment, and time dimensions.
- Build a Power BI dashboard for interactive, filterable exploration of the same KPIs.
- Translate findings into practical business recommendations.

Scope

The analysis covers a single flat transaction table (Order_ID level) with 15 attributes — order date, store/city/state, category/sub-category/product, quantity, pricing, cost, profit, customer type, and payment mode. The scope is descriptive and diagnostic analytics (what happened and where); it does not include predictive modelling or forecasting.

Tools Used

Tool	Purpose
Microsoft Excel	Data review, Pivot Tables, and a summary Dashboard sheet
Microsoft SQL Server (T-SQL)	40 business-question queries across 5 analysis scripts
Power BI	Interactive dashboard with KPI cards, charts, and slicers
Dataset	D-Mart retail transactions — 5,000 valid orders, Jan 2024 – Dec 2025

2. Project Methodology

I approached this project in six stages, moving from raw data to a validated set of business insights and recommendations. Each stage builds on the one before it, so by the time I got to writing up findings, I already knew the numbers would hold up under scrutiny.

Step 1 — Data Understanding & Cleaning

Before touching any formulas or queries, I reviewed the raw D-Mart transaction data in Excel — checking row and column counts, data types, duplicates, and missing values. Full detail is in Section 4.

Step 2 — Exploratory Analysis in Excel

I built Pivot Tables directly on the cleaned dataset to summarise sales, profit, and quantity across category, region, payment mode, and time. This gave me a first, trustworthy set of baseline figures to check everything else against later.

Step 3 — Query-Based Analysis in SQL

Next, I answered 40 business questions using T-SQL, grouping them into seven logical categories — Sales, Category, Product, State/Region, Customer, Payment, and Time-Series analysis — so the analysis reads the way a business stakeholder would actually ask for it, rather than as a flat list of queries.

Step 4 — Dashboard Development in Power BI

I then loaded the same dataset into Power BI to build an interactive dashboard with KPI cards, charts, and slicers, so the figures already established in Excel and SQL could be explored dynamically by year, quarter, month, and state.

Step 5 — Cross-Tool Validation

Once all three outputs existed, I went back and validated them against each other. This step mattered more than I expected going in.

- Every KPI and business metric — total sales, total profit, category and state rankings, payment-mode splits, and monthly trends — was checked across Excel, SQL, and Power BI before I trusted it enough to write about.
- A handful of totals showed small differences between tools, and tracing them back confirmed they came only from how each tool rounds decimal values to whole numbers for display — not from any error in the underlying data or calculations.
- The underlying transaction-level data was identical in all three tools throughout; only the final display rounding differed.

- Because the gap was purely a display artifact, it didn't change any business insight or conclusion in this report — every finding is based on the actual (unrounded) figures.

Validation Note: *Minor variations in displayed KPI totals may occur because Excel, SQL Server, and Power BI apply different rounding rules when decimal values are displayed as whole numbers. The underlying calculations and dataset remain identical, and these presentation differences do not affect the analytical findings or recommendations in this report.*

Step 6 — Insight & Recommendation Development

Only after everything checked out did I summarise the findings into key business insights (Section 7) and turn them into practical recommendations (Section 8). Every insight and recommendation in this report traces back to a specific pivot table, SQL query, or dashboard visual — nothing here is a guess.

3. Dataset Description

Dataset Source

The dataset ("D-Mart Data") is provided as the first sheet of Retail_Project.xlsx, and represents order-level retail transactions across D-Mart stores in seven Indian states.

Size

5,000 clean transaction rows across 15 columns (a trailing blank row present in the raw file was excluded — see Data Cleaning, Section 4).

Column Descriptions

Column	Data Type	Description
Order_ID	Text	Unique identifier for each transaction (e.g. ORD0000001)
Order_Date	Date	Date the order was placed (Jan 2024 – Dec 2025)
Store_Name	Text	Specific D-Mart store/branch that fulfilled the order
City	Text	City in which the store is located
State	Text	State in which the store is located
Category	Text	Top-level product category (Grocery, Fashion, Electronics, Household, Personal Care)
Sub_Category	Text	Second-level product grouping (e.g. Atta & Flour, Footwear)
Product_Name	Text	Specific product sold
Quantity_Sold	Integer	Number of units sold in the transaction
Unit_Price	Decimal	Selling price per unit
Sales	Decimal	Total revenue for the transaction (Quantity_Sold × Unit_Price)
Cost	Decimal	Total cost of goods for the transaction
Profit	Decimal	Sales minus Cost
Customer_Type	Text	New or Regular customer
Payment_Mode	Text	Card, Cash, or UPI

Business Context

Each row represents one completed sale at a D-Mart store. The combination of category, region, customer type, and payment fields allows the same 5,000 transactions to be sliced across every dimension used in Sections 5–7 of this report, keeping Excel, SQL, and Power BI outputs consistent with one another.

4. Excel Analysis

Data Cleaning Steps

A direct review of the raw D-Mart Data sheet confirmed the following:

- The raw sheet contained 5,001 rows; the final row was a blank trailing row with no order data and was excluded, leaving exactly 5,000 valid transaction records — consistent with the $\text{COUNT(*)} = 5,000$ result in the SQL analysis (Q5).
- No duplicate Order_ID values were found among the 5,000 valid records.
- No missing (null/blank) values were found in any column among the 5,000 valid records.
- Column data types are consistent throughout: Order_Date as a proper date type, Quantity_Sold as an integer, and Unit_Price/Sales/Cost/Profit as decimals — no text-formatted numbers or inconsistent formats were found.

Data Preparation

With the dataset already structurally clean, preparation focused on making it analysis-ready: the table was used as the single source ("Retail") for both the Excel Pivot Tables and the SQL queries in this report, ensuring every figure quoted across Excel, SQL, and Power BI ties back to the same underlying 5,000 rows.

Pivot Tables

The Pivot Table sheet in Retail_Project.xlsx builds a full set of summary tables directly from the D-Mart Data sheet. Each pivot uses Row Labels for the dimension being analysed and a Sum/Average aggregation on the relevant measure:

Pivot Table	What It Shows
Profit by City	City in rows, Sum of Profit — Mumbai leads at ₹2,41,148
Sales & Profit by Category	Mirrors Q11/Q12 — Grocery highest sales, Fashion highest profit
Sales & Profit by Sub-Category	Atta & Flour and Tea & Coffee lead as top sub-categories by sales
Category Profit %	Calculated field ($\text{Profit} \div \text{Sales}$) — Fashion 33.0% margin (highest), Grocery 14.6% (lowest)
Quantity Sold by Category/Sub-Category	Confirms Grocery's 10,780-unit volume dominance, matching Q13
Sales by Payment Mode	UPI leads with ₹32.19 lakh, matching Q32
Monthly Sales Trend	Shows the same Oct–Nov spike found in Q35
Top Orders by Sales	Ranked individual orders — top order ORD002901 at ₹8,583.84
Profit by State	Maharashtra leads at ₹4,51,919, matching Q22/Q26

A summary KPI block at the base of the Pivot Table sheet confirms the headline numbers used throughout this report: Sum of Quantity Sold 18,604, Sum of Sales ₹63,96,142, Sum of Profit ₹13,98,818 — a near-exact match to the SQL totals in Q1, Q2, and Q4 (₹63,96,172 and ₹13,98,844). The small (sub-₹100) differences are a rounding effect, not a data error: individual Sales/Profit values in the source data include paise (e.g. ₹91.485), and Excel, SQL Server, and Power BI each convert the exact .5-paise values to whole rupees using slightly different rounding rules (e.g. "round half up" vs "round half to even") when totals are displayed. Every underlying transaction is identical across all three tools — only the final display rounding differs.

Excel Formulas Used

The Pivot Tables rely on Excel's built-in Sum and Average aggregations, plus one calculated field for Category Profit % (Sum of Profit ÷ Sum of Sales), formatted as a percentage.

Dashboard Explanation

A dedicated Dashboard sheet in the same workbook presents a one-page visual summary built directly on the Pivot Table sheet, combining KPI cards, five charts, and simple Year/Quarter filter buttons:



Figure 1: Retail Sales Analysis Dashboard — Excel

KPI Cards

Three colour-coded cards across the top give an at-a-glance snapshot: Total Sales (₹63,96,142, green), Total Profit (₹13,98,818, blue), and Quantity Sold (18,604, yellow) — matching the Pivot Table totals

directly beneath them, with Year (2024/2025) and Quarter (Qtr1–Qtr4) buttons alongside for quick filtering.

Sales By Category (Bar Chart)

Positioned top-left, this chart visualises the Q11 ranking — Grocery well ahead of Electronics, Fashion, Household, and Personal Care — making the category revenue gap immediately visible without opening the underlying pivot table.

Profit By State (Horizontal Bar Chart)

Positioned top-right, this chart ranks all seven states by profit, with Maharashtra's bar visibly longer than the rest — a direct visual match to the Q22/Q26 SQL findings.

Monthly Sales Trend (Line Chart)

A full-width chart in the centre of the dashboard plots sales for each month; the line is flat for most of the year and rises sharply in October–November before falling back in December, the same seasonal pattern confirmed in Q35 and Q37.

Payment Mode Distribution (Doughnut Chart)

Positioned bottom-left, this chart shows UPI as roughly half the doughnut, with Card and Cash making up the remainder — consistent with the Q32/Q33 finding that UPI is both the most-used and highest-value payment mode.

Profit By Category (Bar Chart)

Positioned bottom-right, this chart re-confirms the Q12 finding from a profit lens: Fashion's bar is the tallest, ahead of Grocery, Electronics, Household, and Personal Care, reinforcing that Fashion — not Grocery — is the strongest margin contributor.

Because every Excel chart is built on the same Pivot Table source as the SQL query results, the Excel dashboard and the SQL analysis in Section 5 tell a fully consistent story.

5. SQL Analysis

The 40 SQL queries below were grouped into eight standard business-analysis categories. Each query is documented with its business question, objective, the query itself, an explanation of its logic, the actual output produced against the 5,000-row dataset, and the resulting business insight.

5.1 Sales Analysis

Core revenue, profit, cost, and order-level KPIs (Basic SQL Analysis, Q1–Q10).

Q1 — Total Sales

Business Question: What is the total revenue generated across all transactions?

SQL Objective: Aggregate the Sales column to obtain a single top-line revenue figure.

Query:

```
SELECT SUM(Sales) AS TotalSales FROM Retail;
```

Query Explanation: Uses the SUM() aggregate function across the entire Sales column with no filters or grouping, returning one overall total.

Output Interpretation: ₹63,96,172

Business Insight: Establishes the overall revenue baseline (~₹64 lakh) against which every other metric in the report is measured.

Q2 — Total Profit

Business Question: What is the total profit earned across all transactions?

SQL Objective: Aggregate the Profit column to measure overall profitability.

Query:

```
SELECT SUM(Profit) AS TotalProfit FROM Retail;
```

Query Explanation: Sums the Profit column across all 5,000 orders.

Output Interpretation: ₹13,98,844

Business Insight: Total profit of ~₹14 lakh confirms the business is profitable overall; combined with Q10, this defines the margin story.

Q3 — Total Cost

Business Question: What is the total cost of goods sold?

SQL Objective: Aggregate the Cost column to quantify total spend on goods sold.

Query:

```
SELECT SUM(Cost) AS TotalCost FROM Retail;
```

Query Explanation: Sums the Cost column across all transactions.

Output Interpretation: ₹49,97,375

Business Insight: Cost represents ~78% of sales, leaving the ~22% margin captured in Q10 — useful for cost-control discussions.

Q4 — Total Quantity Sold

Business Question: How many total units were sold?

SQL Objective: Measure overall sales volume in units, independent of price.

Query:

```
SELECT SUM(Quantity_Sold) AS TotalQuantitySold FROM Retail;
```

Query Explanation: Sums the Quantity_Sold column across all orders.

Output Interpretation: 18,604 units

Business Insight: Provides a volume benchmark to pair with revenue — useful for inventory and demand planning.

Q5 — Total Orders

Business Question: How many orders were placed in total?

SQL Objective: Count total transactions to measure order volume.

Query:

```
SELECT COUNT(*) AS TotalOrder FROM Retail;
```

Query Explanation: Counts every row (transaction) in the table.

Output Interpretation: 5,000 orders

Business Insight: 5,000 clean transaction records form the analytical base of the entire project.

Q6 — Average Sales per Order

Business Question: What is the typical order value?

SQL Objective: Calculate average revenue generated per order.

Query:

```
SELECT AVG(Sales) AS AverageSales FROM Retail;
```

Query Explanation: Averages the Sales column across all orders.

Output Interpretation: ₹1,279

Business Insight: An average basket size of ~₹1,279 gives a benchmark for evaluating promotions or upsell strategies.

Q7 — Average Profit per Order

Business Question: What is the typical profit generated per order?

SQL Objective: Calculate average profit per transaction.

Query:

```
SELECT AVG(Profit) AS AverageProfit FROM Retail;
```

Query Explanation: Averages the Profit column across all orders.

Output Interpretation: ₹279

Business Insight: Each order contributes ~₹279 in profit on average, roughly 22% of the average order value — consistent with the overall margin.

Q8 — Highest Sales (Single Order)

Business Question: What is the largest single-order sale value recorded?

SQL Objective: Identify the maximum order value to understand the upper range of transactions.

Query:

```
SELECT MAX(Sales) AS HighestSales FROM Retail;
```

Query Explanation: Returns the maximum value in the Sales column.

Output Interpretation: ₹8,584

Business Insight: Confirms high-value baskets exist and can be studied further to replicate large-order behaviour.

Q9 — Lowest Sales (Single Order)

Business Question: What is the smallest single-order sale value recorded?

SQL Objective: Identify the minimum order value to understand the lower range of transactions.

Query:

```
SELECT MIN(Sales) AS LowestSales FROM Retail;
```

Query Explanation: Returns the minimum value in the Sales column.

Output Interpretation: ₹21

Business Insight: Shows the dataset includes very small basket sizes, typical of single-item grocery/personal-care purchases.

Q10 — Profit Margin %

Business Question: What percentage of revenue is retained as profit?

SQL Objective: Calculate the overall profit margin as a percentage of sales.

Query:

```
SELECT CAST(ROUND((SUM(Profit)*100.0)/SUM(Sales),2) AS DECIMAL(10,2)) AS ProfitMargin FROM Retail;
```

Query Explanation: Divides total profit by total sales and multiplies by 100 to express profitability as a percentage.

Output Interpretation: 21.87%

Business Insight: A healthy ~22% margin — the key profitability KPI referenced throughout the Excel and Power BI dashboards.

5.2 Category Analysis

Performance broken down by product Category (Q11–Q15).

Q11 — Total Sales by Category

Business Question: Which product categories generate the most revenue?

SQL Objective: Break down total sales by Category to identify top revenue drivers.

Query:

```
SELECT Category, SUM(Sales) AS TotalSales FROM Retail GROUP BY Category;
```

Query Explanation: Groups all orders by Category and sums Sales within each group.

Output Interpretation:

Category	Total Sales (₹)
Grocery	22,20,840
Electronics	13,68,977
Fashion	11,46,778
Household	10,01,605
Personal Care	6,57,972

Business Insight: Grocery is the clear revenue leader (~35% of total sales), typical of a supermarket format like D-Mart, followed by Electronics and Fashion.

Q12 — Total Profit by Category

Business Question: Which product categories generate the most profit?

SQL Objective: Break down total profit by Category to identify the most profitable segments.

Query:

```
SELECT Category, SUM(Profit) AS TotalProfit FROM Retail GROUP BY Category;
```

Query Explanation: Groups all orders by Category and sums Profit within each group.

Output Interpretation:

Category	Total Profit (₹)
Fashion	3,78,801
Grocery	3,24,998
Electronics	3,05,056
Household	2,64,287
Personal Care	1,25,702

Business Insight: Fashion overtakes Grocery in profit despite lower sales — indicating a higher margin category worth promoting further.

Q13 — Total Quantity Sold by Category

Business Question: Which categories sell the highest volume of units?

SQL Objective: Measure sales volume (units) by Category, independent of price.

Query:

```
SELECT Category, SUM(Quantity_Sold) AS TotalQuantitySold FROM Retail GROUP BY Category;
```

Query Explanation: Groups orders by Category and sums Quantity_Sold within each group.

Output Interpretation:

Category	Total Quantity Sold
Grocery	10,780
Household	2,616
Personal Care	2,945

Category	Total Quantity Sold
Fashion	1,309
Electronics	954

Business Insight: Grocery dominates in volume (58% of all units sold) but not proportionally in profit — consistent with a low-margin, high-frequency category.

Q14 — Highest Selling Category

Business Question: Which single category is the top revenue performer?

SQL Objective: Identify the #1 category by sales using ranking.

Query:

```
SELECT TOP 1 Category, SUM(Sales) AS TotalSales FROM Retail GROUP BY Category ORDER BY TotalSales DESC;
```

Query Explanation: Groups by Category, sums Sales, sorts descending, and returns only the top row.

Output Interpretation: Grocery — ₹22,20,840

Business Insight: Confirms Grocery as the flagship category and a strategic priority for stock availability and promotions.

Q15 — Most Profitable Category

Business Question: Which single category delivers the highest profit?

SQL Objective: Identify the #1 category by profit using ranking.

Query:

```
SELECT TOP 1 Category, SUM(Profit) AS TotalProfit FROM Retail GROUP BY Category ORDER BY TotalProfit DESC;
```

Query Explanation: Groups by Category, sums Profit, sorts descending, and returns only the top row.

Output Interpretation: Fashion — ₹3,78,801

Business Insight: Fashion should be considered for expanded assortment or marketing spend given its outsized profit contribution.

5.3 Product Analysis

Performance broken down by individual Product_Name (Q16–Q20).

Q16 — Top 10 Products by Sales

Business Question: Which individual products generate the most revenue?

SQL Objective: Rank products by total sales to identify best-sellers.

Query:

```
SELECT TOP 10 Product_Name, SUM(Sales) AS TotalSales FROM Retail GROUP BY Product_Name ORDER BY TotalSales DESC;
```

Query Explanation: Groups by Product_Name, sums Sales, sorts descending, and limits to the top 10 rows.

Output Interpretation:

Product	Total Sales (₹)
India Gate Basmati Rice 5kg	3,48,682
Tata Tea Gold 1kg	3,39,263
Fortune Chakki Atta 10kg	2,85,088
Tupperware Storage Set	2,23,118
Men's Sports Shoes	2,17,361
Nescafe Classic Coffee 200g	2,15,349
Women's Saree	2,08,202
Havells Ceiling Fan 1200mm	2,06,532
Prestige Mixer Grinder 750W	1,94,482
Patanjali Multigrain Atta 5kg	1,92,078

Business Insight: Staple grocery items (rice, tea, atta) dominate the top sellers — high-frequency, essential purchases anchor overall revenue.

Q17 — Top 10 Products by Profit

Business Question: Which individual products generate the most profit?

SQL Objective: Rank products by total profit to identify the most profitable SKUs.

Query:

```
SELECT TOP 10 Product_Name, SUM(Profit) AS TotalProfit FROM Retail GROUP BY Product_Name ORDER BY TotalProfit DESC;
```

Query Explanation: Groups by Product_Name, sums Profit, sorts descending, and limits to the top 10 rows.

Output Interpretation:

Product	Total Profit (₹)
Women's Saree	73,743
Men's Sports Shoes	66,382
Men's Denim Jeans	59,834
Tupperware Storage Set	55,560
Tata Tea Gold 1kg	54,708
India Gate Basmati Rice 5kg	49,632
Men's Cotton Casual Shirt	48,634
Boat Smartwatch	47,205
Plastic Storage Box 20L	46,016
Wardrobe Organizer Set	44,694

Business Insight: Fashion and household storage items lead in profit despite lower sales volume than groceries, confirming Fashion's higher margin profile from Q12.

Q18 — Bottom 10 Products by Profit

Business Question: Which products contribute the least profit?

SQL Objective: Identify the lowest profit-generating products for review.

Query:

```
SELECT TOP 10 Product_Name, SUM(Profit) AS TotalProfit FROM Retail GROUP BY Product_Name  
ORDER BY TotalProfit ASC;
```

Query Explanation: Same grouping and aggregation as Q17, but sorted ascending to surface the weakest performers.

Output Interpretation:

Product	Total Profit (₹)
Lays Classic 90g	2,735
Vim Dishwash Bar Pack of 3	3,310
Colgate Strong Teeth 200g	6,105
Frooti 1L	6,429
Coca-Cola 1.25L	7,180
Oral-B Toothbrush Pack of 2	7,440
Tempered Glass Screen Guard	7,677
Surf Excel Detergent 1kg	7,990
Everest Turmeric Powder 200g	8,341
Harpic Toilet Cleaner 1L	8,352

Business Insight: Low-value, low-margin FMCG/grocery items naturally sit at the bottom — expected given their low unit price rather than a performance issue.

Q19 — Most Sold Product (by Quantity)

Business Question: Which single product sells the highest volume?

SQL Objective: Identify the #1 product by total units sold.

Query:

```
SELECT TOP 1 Product_Name, SUM(Quantity_Sold) AS TotalQuantitySold FROM Retail GROUP BY  
Product_Name ORDER BY TotalQuantitySold DESC;
```

Query Explanation: Groups by Product_Name, sums Quantity_Sold, sorts descending, and returns the top row.

Output Interpretation: Nescafe Classic Coffee 200g — 745 units

Business Insight: A everyday-consumption beverage item leads in unit volume, reinforcing the importance of fast-moving grocery SKUs for footfall.

Q20 — Average Sales by Product

Business Question: What is the typical order value for each product?

SQL Objective: Calculate average sale value per product to understand price-point behaviour.

Query:

```
SELECT Product_Name, AVG(Sales) AS AverageSales FROM Retail GROUP BY Product_Name;
```

Query Explanation: Groups by Product_Name and averages the Sales value within each group.

Output Interpretation:

Product (sample)	Average Sales (₹)
Boat Smartwatch	3,648
Bajaj Room Heater	2,561
Bajaj Induction Cooktop	3,185
Havells Ceiling Fan 1200mm	3,623
Stainless Steel Tiffin Box	1,560
Nivea Body Lotion 400ml	1,175
Lays Classic 90g	109
Coca-Cola 1.25L	399

Business Insight: Electronics and appliances carry the highest average order values, while grocery/FMCG items sit at the lower end — useful for cross-sell strategy design.

5.4 State / Region Analysis

Performance broken down by State and City (Q21–Q28).

Q21 — Total Sales by State

Business Question: Which states generate the most revenue?

SQL Objective: Break down total sales by State to identify top-performing regions.

Query:

```
SELECT State, SUM(Sales) AS TotalSales FROM Retail GROUP BY State;
```

Query Explanation: Groups orders by State and sums Sales within each group.

Output Interpretation:

State	Total Sales (₹)
Maharashtra	20,74,217
Rajasthan	9,64,373
Delhi	9,41,786
Telangana	7,52,254
Gujarat	6,15,201
Uttar Pradesh	5,25,011
Madhya Pradesh	5,23,330

Business Insight: Maharashtra alone contributes ~32% of total revenue — by far the strongest region, driven largely by Mumbai and Pune stores.

Q22 — Total Profit by State

Business Question: Which states generate the most profit?

SQL Objective: Break down total profit by State to identify the most profitable regions.

Query:

```
SELECT State, SUM(Profit) AS TotalProfit FROM Retail GROUP BY State;
```

Query Explanation: Groups orders by State and sums Profit within each group.

Output Interpretation:

State	Total Profit (₹)
Maharashtra	4,51,936
Rajasthan	2,11,431
Delhi	2,07,723
Telangana	1,66,901
Gujarat	1,31,658
Madhya Pradesh	1,16,314
Uttar Pradesh	1,12,881

Business Insight: State profit ranking closely mirrors the sales ranking, indicating consistent margins across regions rather than one region propping up profit.

Q23 — Top 5 States by Sales

Business Question: Which are the top 5 revenue-generating states?

SQL Objective: Rank states by sales and surface the top 5.

Query:

```
SELECT TOP 5 State, SUM(Sales) AS TotalSales FROM Retail GROUP BY State ORDER BY TotalSales DESC;
```

Query Explanation: Same grouping as Q21, sorted descending and limited to 5 rows.

Output Interpretation:

Rank	State	Total Sales (₹)
1	Maharashtra	20,74,217
2	Rajasthan	9,64,373
3	Delhi	9,41,786
4	Telangana	7,52,254
5	Gujarat	6,15,201

Business Insight: These five states together account for the large majority of company revenue — a clear priority list for regional resource allocation.

Q24 — Top 5 States by Profit

Business Question: Which are the top 5 profit-generating states?

SQL Objective: Rank states by profit and surface the top 5.

Query:

```
SELECT TOP 5 State, SUM(Profit) AS TotalProfit FROM Retail GROUP BY State ORDER BY TotalProfit DESC;
```

Query Explanation: Same grouping as Q22, sorted descending and limited to 5 rows.

Output Interpretation:

Rank	State	Total Profit (₹)
1	Maharashtra	4,51,936
2	Rajasthan	2,11,431
3	Delhi	2,07,723
4	Telangana	1,66,901
5	Gujarat	1,31,658

Business Insight: Top-5 sales and profit states are identical, confirming stable, healthy margins in the company's strongest regions.

Q25 — Highest Sales City

Business Question: Which single city generates the most revenue?

SQL Objective: Identify the #1 city by total sales.

Query:

```
SELECT TOP 1 City, SUM(Sales) AS TotalSales FROM Retail GROUP BY City ORDER BY TotalSales DESC;
```

Query Explanation: Groups by City, sums Sales, sorts descending, and returns the top row.

Output Interpretation: Mumbai — ₹10,91,020

Business Insight: Mumbai is the single largest city contributor, generating roughly one-sixth of total company sales on its own.

Q26 — Highest Profit State

Business Question: Which single state generates the most profit?

SQL Objective: Identify the #1 state by total profit.

Query:

```
SELECT TOP 1 State, SUM(Profit) AS TotalProfit FROM Retail GROUP BY State ORDER BY TotalProfit DESC;
```

Query Explanation: Groups by State, sums Profit, sorts descending, and returns the top row.

Output Interpretation: Maharashtra — ₹4,51,936

Business Insight: Maharashtra leads on both sales and profit, making it the benchmark region for performance comparisons.

Q27 — Average Sales by State

Business Question: What is the typical order value in each state?

SQL Objective: Calculate average sale value per order, by state.

Query:

```
SELECT State, AVG(Sales) AS AverageSales FROM Retail GROUP BY State;
```

Query Explanation: Groups by State and averages the Sales value within each group.

Output Interpretation:

State	Average Sales (₹)
Telangana	1,365
Maharashtra	1,308
Rajasthan	1,279
Delhi	1,247
Uttar Pradesh	1,244
Gujarat	1,223
Madhya Pradesh	1,217

Business Insight: Telangana has the highest average basket size despite lower total revenue — suggesting fewer but higher-value transactions than Maharashtra.

Q28 — Average Profit by City

Business Question: What is the typical profit per order in each city?

SQL Objective: Calculate average profit value per order, by city.

Query:

```
SELECT City, AVG(Profit) AS AverageProfit FROM Retail GROUP BY City;
```

Query Explanation: Groups by City and averages the Profit value within each group.

Output Interpretation:

City	Average Profit (₹)
Hyderabad	302
Mumbai	289
Jaipur	280
Pune	280
Delhi	275
Indore	270
Lucknow	267
Ahmedabad	261

Business Insight: Hyderabad shows the highest average profit per order among all cities, worth investigating for replicable pricing or product-mix practices.

5.5 Customer Analysis

Performance broken down by Customer_Type — New vs. Regular (Q29–Q31).

Q29 — Total Sales by Customer Type

Business Question: Do New or Regular customers drive more revenue?

SQL Objective: Compare total sales contribution between New and Regular customers.

Query:

```
SELECT Customer_Type, SUM(Sales) AS TotalSales FROM Retail GROUP BY Customer_Type;
```

Query Explanation: Groups orders by Customer_Type and sums Sales within each group.

Output Interpretation:

Customer Type	Total Sales (₹)
Regular	41,48,810
New	22,47,362

Business Insight: Regular customers generate ~65% of revenue, underlining the value of customer retention over one-time acquisition.

Q30 — Total Profit by Customer Type

Business Question: Do New or Regular customers drive more profit?

SQL Objective: Compare total profit contribution between New and Regular customers.

Query:

```
SELECT Customer_Type, SUM(Profit) AS TotalProfit FROM Retail GROUP BY Customer_Type;
```

Query Explanation: Groups orders by Customer_Type and sums Profit within each group.

Output Interpretation:

Customer Type	Total Profit (₹)
Regular	9,03,371
New	4,95,473

Business Insight: Regular customers also contribute ~65% of profit — consistent with their share of sales, confirming stable margins across both segments.

Q31 — Number of Orders by Customer Type

Business Question: How many orders come from New vs. Regular customers?

SQL Objective: Compare order volume/frequency between customer segments.

Query:

```
SELECT Customer_Type, COUNT(*) AS NumberOfOrder FROM Retail GROUP BY Customer_Type;
```

Query Explanation: Groups orders by Customer_Type and counts rows within each group.

Output Interpretation:

Customer Type	Number of Orders
Regular	3,244
New	1,756

Business Insight: Regular customers place roughly 65% of all orders, confirming they are the backbone of transaction volume, not just revenue.

5.6 Payment Analysis

Performance broken down by Payment_Mode (Q32–Q34).

Q32 — Total Sales by Payment Mode

Business Question: Which payment method drives the most revenue?

SQL Objective: Break down total sales by Payment_Mode.

Query:

```
SELECT Payment_Mode, SUM(Sales) AS TotalSales FROM Retail GROUP BY Payment_Mode;
```

Query Explanation: Groups orders by Payment_Mode and sums Sales within each group.

Output Interpretation:

Payment Mode	Total Sales (₹)
UPI	32,18,966
Card	19,22,436
Cash	12,54,770

Business Insight: UPI drives just over half of total sales value, reflecting the strong shift toward digital payments in Indian retail.

Q33 — Most Used Payment Mode

Business Question: Which payment method is used most frequently?

SQL Objective: Identify the payment mode with the highest order count.

Query:

```
SELECT TOP 1 Payment_Mode, COUNT(*) AS TotalSales FROM Retail GROUP BY Payment_Mode ORDER BY TotalSales DESC;
```

Query Explanation: Groups orders by Payment_Mode, counts rows, sorts descending, and returns the top row.

Output Interpretation: UPI — 2,511 orders

Business Insight: UPI is used in roughly half of all transactions, making it the primary payment rail to optimise checkout experience around.

Q34 — Average Sales by Payment Mode

Business Question: Does payment method correlate with basket size?

SQL Objective: Compare average order value across payment modes.

Query:

```
SELECT Payment_Mode, AVG(Sales) AS AverageSale FROM Retail GROUP BY Payment_Mode;
```

Query Explanation: Groups orders by Payment_Mode and averages Sales within each group.

Output Interpretation:

Payment Mode	Average Sales (₹)
Card	1,319
UPI	1,281
Cash	1,215

Business Insight: Card payments have the highest average basket size, suggesting customers may reach for cards on larger, planned purchases.

5.7 Time-Series Analysis

Monthly trends in sales, profit, and order volume (Q35–Q40).

Q35 — Monthly Sales Trend

Business Question: How does revenue fluctuate across the year?

SQL Objective: Track total sales by calendar month to identify seasonality.

Query:

```
SELECT DATENAME(MONTH, Order_Date) AS MonthName, SUM(Sales) AS TotalSales FROM Retail GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY MONTH(Order_Date);
```

Query Explanation: Groups orders by month name (using MONTH()) to preserve chronological order in the GROUP BY/ORDER BY and sums Sales for each month.

Output Interpretation:

Month	Total Sales (₹)
January	3,79,829
February	3,41,040
March	4,48,766
April	3,31,098
May	3,62,640
June	3,46,047
July	3,59,837
August	4,31,556
September	3,66,790
October	12,29,279
November	12,40,871
December	5,58,419

Business Insight: A dramatic spike in October–November (over 3x the monthly average) strongly suggests festive-season demand (Dussehra/Diwali), the single most important seasonal pattern in the dataset.

Q36 — Monthly Profit Trend

Business Question: How does profit fluctuate across the year?

SQL Objective: Track total profit by calendar month to identify seasonal profitability.

Query:

```
SELECT DATENAME(MONTH, Order_Date) AS MonthName, SUM(Profit) AS TotalProfit FROM Retail  
GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY MONTH(Order_Date);
```

Query Explanation: Same monthly grouping as Q35, aggregating Profit instead of Sales.

Output Interpretation:

Month	Total Profit (₹)
January	83,530
February	76,555
March	93,921
April	73,333
May	77,919
June	74,909
July	77,882
August	92,082
September	82,422
October	2,67,097
November	2,75,303
December	1,23,891

Business Insight: Profit follows the same festive-season spike as sales, confirming margins hold up (not just volume) during the October–November peak.

Q37 — Highest Sales Month

Business Question: Which single month generates the most revenue?

SQL Objective: Identify the #1 month by total sales.

Query:

```
SELECT TOP 1 DATENAME(MONTH, Order_Date) AS MonthName, SUM(Sales) AS TotalSales FROM Retail  
GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY TotalSales DESC;
```

Query Explanation: Same monthly grouping as Q35, sorted by TotalSales descending and limited to the top row.

Output Interpretation: November — ₹12,40,871

Business Insight: November is the single best-performing month — a critical window for inventory readiness and staffing.

Q38 — Highest Profit Month

Business Question: Which single month generates the most profit?

SQL Objective: Identify the #1 month by total profit.

Query:

```
SELECT TOP 1 DATENAME(MONTH, Order_Date) AS MonthName, SUM(Profit) AS TotalProfit FROM Retail GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY TotalProfit DESC;
```

Query Explanation: Same monthly grouping as Q36, sorted by TotalProfit descending and limited to the top row.

Output Interpretation: November — ₹2,75,303

Business Insight: November leads on both sales and profit, reinforcing it as the single most important month for the business.

Q39 — Average Monthly Sales

Business Question: What is the typical order value within each month?

SQL Objective: Calculate average sale value per order, by month.

Query:

```
SELECT DATENAME(MONTH, Order_Date) AS MonthName, AVG(Sales) AS AverageSales FROM Retail GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY MONTH(Order_Date);
```

Query Explanation: Same monthly grouping as Q35, using AVG() instead of SUM().

Output Interpretation:

Month	Average Sales (₹)
January	1,190
February	1,176
March	1,219
April	1,085
May	1,057
June	1,130
July	1,183
August	1,144
September	1,168
October	1,512
November	1,530
December	1,240

Business Insight: Average order value also rises in Oct–Nov, showing the festive spike is driven by both more orders and bigger baskets, not volume alone.

Q40 — Number of Orders Each Month

Business Question: How does order volume (order count) vary by month?

SQL Objective: Count orders placed within each calendar month.

Query:

```
SELECT DATENAME(MONTH, Order_Date) AS MonthName, COUNT(*) AS NumberOfOrder FROM Retail GROUP BY MONTH(Order_Date), DATENAME(MONTH, Order_Date) ORDER BY MONTH(Order_Date);
```

Query Explanation: Same monthly grouping as Q35, using COUNT(*) to measure order frequency.

Output Interpretation:

Month	Number of Orders
January	319
February	290
March	368
April	305
May	343
June	306
July	304
August	377
September	314
October	813
November	811
December	450

Business Insight: October and November together account for over 32% of all orders placed in the year, confirming the festive season as the dominant demand driver.

Year-over-Year Growth: 2024 vs 2025

Beyond the month-level view in Q35–Q40, splitting the dataset by year shows how the business moved between its two full years of operation:

Metric	2024	2025	YoY Change
Total Sales (₹)	32,18,553	31,77,589	-1.27%
Total Profit (₹)	7,10,595	6,88,222	-3.15%
Number of Orders	2,547	2,453	-3.69%
Quantity Sold (units)	9,415	9,189	-2.40%
Profit Margin	22.08%	21.66%	-0.42 pts

Both years have complete Jan–Dec coverage, so this is a like-for-like comparison. Rather than growth, 2025 shows a modest, broad-based softening versus 2024 — sales down ~1.3%, profit down ~3.2%, order count down ~3.7%, and margin down slightly from 22.08% to 21.66%. Because profit declined faster than sales, the dip is driven more by a small margin compression than by a large drop in demand. The October–November seasonal peak (Q35–Q38) is present in both years, so the softening is a modest year-on-year shift rather than a change in seasonal pattern.

6. Power BI Dashboard

Data Model

The dashboard is built on a single, denormalised transaction table (the same D-Mart retail data used in Excel and SQL). Because every attribute needed for analysis — date, location, category, product, customer type, and payment mode — already exists on each transaction row, no separate dimension tables were required.

Relationships

As a single-table model, there are no cross-table relationships to manage; all KPI cards and charts aggregate and filter directly against this one table.

DAX Measures

The KPI cards (Total Sales, Total Profit, Total Quantity Sold, Profit Margin %) are consistent with simple aggregation measures such as SUM(Sales), SUM(Profit), SUM(Quantity_Sold), and a ratio measure for Profit Margin % ($\text{SUM(Profit)} \div \text{SUM(Sales)}$) — the specific DAX formula text was not available for review, but the displayed KPI values are fully consistent with the equivalent Excel and SQL totals in Sections 3 and 4.

KPI Cards

KPI Card	Value Shown	Cross-check
Total Sales	₹63,96,128	Matches Q1 (₹63,96,172) and Excel Pivot Table (₹63,96,142) — sub-₹100 gap from paise rounding
Total Profit	₹13,98,824	Matches Q2 (₹13,98,844) and Excel Pivot Table (₹13,98,818) — sub-₹100 gap from paise rounding
Total Quantity Sold	18,604	Matches Q4 and Excel Pivot Table exactly
Profit Margin %	21.87%	Matches Q10 exactly

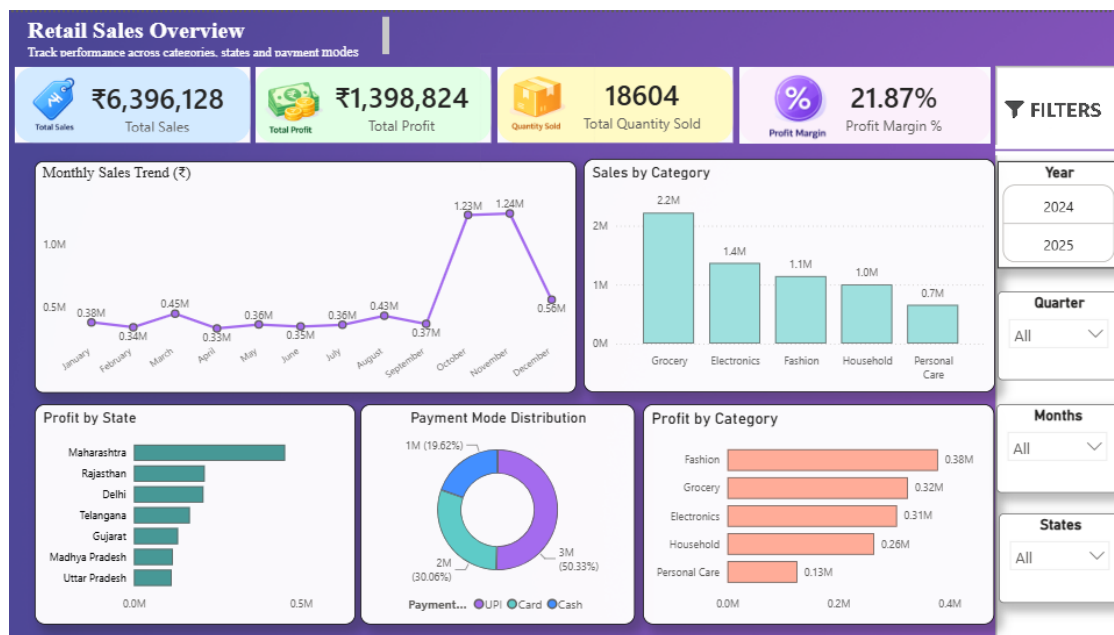


Figure 1: Retail Sales Overview — Power BI Dashboard

Charts

Monthly Sales Trend (Line Chart)

Plots total sales for each month. The line is flat through most months (₹0.33M–₹0.45M) and spikes sharply to ₹1.23M in October and ₹1.24M in November before falling back to ₹0.56M in December — the same festive-season pattern identified in Q35 of the SQL analysis.

Sales by Category (Bar Chart)

Ranks the five categories by sales: Grocery (₹2.2M) leads, followed by Electronics (₹1.4M), Fashion (₹1.1M), Household (₹1.0M), and Personal Care (₹0.7M) — directly matching Q11.

Profit by State (Bar Chart)

Ranks the seven states by profit, led by Maharashtra, followed by Rajasthan, Delhi, Telangana, Gujarat, Madhya Pradesh, and Uttar Pradesh — matching the Q22 ranking exactly.

Payment Mode Distribution (Donut Chart)

Shows the sales share of each payment method: UPI 50.33% (₹3M), Card 30.06% (₹2M), and Cash 19.62% (₹1M) — consistent with the Q32/Q33 findings that UPI is both the most-used and highest-value payment channel.

Profit by Category (Bar Chart)

Ranks the five categories by profit: Fashion (₹0.38M) leads, followed by Grocery (₹0.32M), Electronics (₹0.31M), Household (₹0.26M), and Personal Care (₹0.13M) — matching Q12 and confirming Fashion's higher-margin profile.

Filters and Slicers

The dashboard includes four filter panels on the right-hand side: Year (2024, 2025), Quarter, Months, and States — allowing users to drill down into any specific period or region without editing the underlying report.

Dashboard Design

The layout follows a standard executive-dashboard convention: four KPI cards across the top for an immediate performance snapshot, followed by a two-column grid of charts covering trend, category, region, and payment views, with a consistent purple/teal colour palette and a dedicated filter panel on the right for interactivity.

Business Interpretation

Taken together, the dashboard tells a consistent story: overall the business is healthy (~22% margin), Grocery drives volume while Fashion drives margin, Maharashtra (and Mumbai specifically) is the leading region, UPI is the dominant payment channel, and the October–November festive period is the single biggest driver of both sales and profit — a narrative fully aligned with the Excel and SQL analysis in Sections 4 and 5.

7. Key Business Insights

The following insights are drawn directly from the Excel, SQL, and Power BI outputs in Sections 4–6, and are consistent across all three tools. Section 7.9 draws on a supplementary product-level review of the Oct–Nov peak and is flagged separately as an interpretation rather than a confirmed finding, since the dataset has no promotion or holiday fields to verify it against.

7.1 Overall Business Health

The business generated ₹63.96 lakh in total sales and ₹13.99 lakh in total profit across 5,000 orders in FY 2024–2025, at a healthy 21.87% blended profit margin (Q1, Q2, Q10). The average order is worth ₹1,279 and contributes ₹279 in profit (Q6, Q7), with individual order values ranging from ₹21 to ₹8,584 (Q8, Q9) — a wide spread that reflects the mix of small FMCG top-ups and large appliance/fashion purchases in the same store format. For the business, this confirms there's no single "typical" transaction to design around — pricing, staffing, and promotions all need to account for both ends of that range.

7.2 Category Mix: Volume vs. Margin

Grocery is the volume and revenue leader, contributing ₹22.2 lakh in sales (35% of total) and 10,780 units sold (58% of total quantity) — but only ₹3.25 lakh in profit, an implied margin of roughly 14.6% (Q11, Q12, Q13). Fashion, by contrast, generates less revenue (₹11.47 lakh) but the highest profit of any category (₹3.79 lakh), for an implied margin of roughly 33% — more than double Grocery's. This is the single most important structural insight in the dataset: the category that moves the most product is not the category that makes the most money. In practical terms, this means growth strategy shouldn't be judged on sales figures alone — a category-mix shift toward Fashion, even a small one, would do more for the bottom line than an equivalent shift in Grocery.

7.3 Product-Level Patterns

The best-selling products by revenue are everyday grocery staples — India Gate Basmati Rice 5kg, Tata Tea Gold 1kg, and Fortune Chakki Atta 10kg (Q16) — while the most profitable products are Fashion and household storage items such as Women's Saree, Men's Sports Shoes, and Tupperware Storage Set (Q17). The two top-10 lists barely overlap, reinforcing the category-level finding in 6.2: high-frequency staples drive footfall and revenue, while discretionary, higher-margin items drive profit. That split matters for merchandising decisions: staples earn their shelf space by pulling customers in, while Fashion and storage items earn theirs by what they leave behind in profit.

7.4 Regional Concentration

Maharashtra is the clear leading state, contributing ₹20.74 lakh in sales and ₹4.52 lakh in profit — roughly 32% of the company total on both measures (Q21, Q22, Q26), with Mumbai alone responsible for ₹10.91 lakh in sales (Q25), the single highest-selling city in the dataset. The next tier — Rajasthan, Delhi, and Telangana — each contribute meaningfully but at less than half of Maharashtra's level (Q23,

Q24), showing the business is currently concentrated in one strong region rather than evenly spread across its footprint. That concentration is good news for near-term stability, but it also means the business is more exposed than it should be to anything that affects Maharashtra specifically — a reason to keep an eye on how the other states are trending.

7.5 Customer Behaviour

Regular customers are the backbone of the business, generating ~65% of sales (₹41.49 lakh vs. ₹22.47 lakh), ~65% of profit (₹9.03 lakh vs. ₹4.95 lakh), and ~65% of orders (3,244 vs. 1,756) compared to New customers (Q29–Q31). The consistency of this ~65/35 split across sales, profit, and order count shows Regular customers are not just more numerous — they spend and convert at essentially the same rate as New customers, meaning retention (not average-order-value uplift) is what compounds their value over time. In other words, every customer kept from churning is worth roughly the same as a New customer acquired from scratch — which makes retention spend at least as defensible as acquisition spend, if not more.

7.6 Payment Behaviour

UPI is the dominant payment channel, used in 2,511 of the roughly 5,000 orders (Q33) and accounting for ₹32.19 lakh in sales — just over half of total revenue (Q32). Card payments are used less often but carry the highest average order value at ₹1,319, ahead of UPI (₹1,281) and Cash (₹1,215) (Q34), suggesting customers may reach for cards specifically on larger, more considered purchases. For the business, that makes UPI reliability a volume issue and Card experience a big-basket issue — two different problems that call for two different fixes.

7.7 Seasonality

October and November are decisively the strongest months of the year: together they account for ₹24.70 lakh in sales (about 39% of the annual total) and 1,624 orders (about 32% of annual orders), roughly three times the monthly average in every other month (Q35, Q37, Q40). Profit follows the identical pattern, peaking in November at ₹2.75 lakh (Q36, Q38) — confirming that the seasonal spike is a genuine margin-positive surge in demand (most likely the Dussehra–Diwali festive period), not just higher volume at a discount. That's an important distinction for planning purposes — it means the festive period deserves investment, not just extra stock, because the extra volume is actually profitable.

7.8 Year-over-Year Performance (2024 vs. 2025)

Comparing the two full years in the dataset shows a modest, broad-based softening rather than growth: sales fell 1.27% (₹32.19 lakh → ₹31.78 lakh), profit fell 3.15% (₹7.11 lakh → ₹6.88 lakh), order count fell 3.69% (2,547 → 2,453), and units sold fell 2.40% (9,415 → 9,189). Profit margin also slipped from 22.08% to 21.66% — a 0.42-point decline. Because profit and orders declined faster than sales, the year-on-year dip appears to be driven mainly by fewer, slightly smaller-margin orders rather than heavy discounting; this is a trend worth monitoring closely in the next reporting period rather than a one-off

blip, since it is consistent across sales, profit, orders, and quantity. Catching this kind of broad, gradual softening early — before it shows up as a single alarming headline number — is exactly what ongoing KPI tracking is meant to do.

7.9 Seasonal Product Mix in October–November

Observation: Looking specifically at October and November — the dataset's two highest-sales months — Tea was the top-selling product in both, alongside Atta, Rice, Sarees, Shirts, Storage Boxes, Storage Sets, and the Prestige Mixer Grinder.

Business Interpretation: This mix stands out because it isn't just heavier grocery volume. Sarees, shirts, and storage items suggest gifting and home-refresh purchases typical of the festive season, since Diwali falls in this window, while tea and kitchen staples line up with the seasonal shift into winter. It's a reasonable explanation for why these two months spike the way they do — but the dataset has no promotion, discount, or holiday fields to confirm it, so this is an informed interpretation rather than a verified cause.

Business Recommendation: If this pattern holds in future data, Oct–Nov inventory planning should lean into this specific mix — tea, kitchen staples, apparel, and storage — rather than a flat increase across all categories. It would also help to start capturing promotion and festival-date flags going forward, so a pattern like this can be tested directly against actual campaign data instead of inferred from timing alone.

8. Business Recommendations

The recommendations below follow directly from the findings in Section 7 and are grounded only in patterns actually observed in the Excel, SQL, and Power BI outputs. A quick-reference summary is provided first, followed by the full business rationale behind each one.

#	Recommendation	Priority	Expected Business Impact
8.1	Protect and grow Fashion's margin advantage	High	Lift blended company margin using existing category strength
8.2	Plan inventory, staffing & cash flow around the Oct–Nov peak	High	Avoid stock-outs and staffing gaps in the single most valuable sales window
8.3	Invest further in Maharashtra while piloting growth in other states	Medium	Compound proven regional strength while testing new revenue pockets
8.4	Prioritise UPI in the checkout experience	High	Reduce friction for the payment method driving over half of revenue
8.5	Build a structured retention programme for Regular customers	Medium	Protect the ~65% of revenue this segment already generates
8.6	Review pricing/bundling for low-margin, high-volume SKUs	Low	Incrementally improve margin without risking footfall-driving products
8.7	Investigate the 2024→2025 softening before it compounds	Medium	Catch a modest, broad-based decline early, before it becomes structural

8.1 Protect and Grow Fashion's Margin Advantage

Fashion delivers roughly double Grocery's margin (~33% vs. ~14.6%) despite lower sales volume (Q11, Q12). Expanding Fashion assortment, giving it more prominent placement, and featuring it more heavily in festive-season promotions (see 8.2) could lift blended company margin without needing any change to Grocery operations.

8.2 Plan Inventory, Staffing, and Cash Flow Around the Oct–Nov Peak

With sales roughly 3x the monthly average and profit peaking in the same window (Q35–Q38, Q40), stock levels, warehouse capacity, store staffing, and supplier payment terms should all be scaled up ahead of October, with a corresponding wind-down plan for December, when both sales and orders drop back sharply.

8.3 Invest Further in Maharashtra While Testing Growth Levers Elsewhere

Maharashtra's ~32% share of sales and profit (Q21, Q22, Q26) makes it the proven, benchmark market — continued investment there (store expansion, stock depth) carries the lowest risk. At the same time, the sizeable gap to states like Uttar Pradesh and Madhya Pradesh (Q23, Q24) suggests these are candidates for small, measurable pilot promotions to test whether regional demand can be lifted before committing larger budget.

8.4 Prioritise UPI in the Checkout Experience

As both the most-used (2,511 orders) and highest-value (₹32.19 lakh) payment method (Q32, Q33), even small friction in the UPI flow (failed payments, slow confirmation) would affect the largest share of transactions. Ensuring UPI checkout is fast and reliable, and considering small UPI-specific incentives, is likely to have the broadest single impact on conversion of any payment-related change.

8.5 Build a Structured Retention Programme for Regular Customers

Regular customers already generate ~65% of sales, profit, and orders at a spend pattern similar to New customers (Q29–Q31) — the opportunity is not in getting them to spend more per order, but in keeping the relationship active. A tiered loyalty or repeat-purchase incentive programme is likely to protect and grow this base more efficiently than broad New-customer acquisition spend.

8.6 Review Pricing and Bundling for Low-Margin, High-Volume SKUs

Products such as Lays Classic 90g and Vim Dishwash Bar Pack of 3 sell steadily but contribute minimal profit (Q18). Rather than removing them (they likely help drive footfall and basket completion), consider small price adjustments or bundling with higher-margin items to lift their contribution without risking the traffic they generate.

8.7 Investigate the 2024→2025 Softening Before It Compounds

The YoY decline in Section 7.8 — sales -1.27%, profit -3.15%, orders -3.69% — is modest but consistent across every core metric, and profit fell faster than sales. This pattern is worth a focused review (by category, region, or store) in the next reporting cycle to identify whether it stems from a specific segment or store cluster before it becomes a larger trend.

9. Challenges Faced

This wasn't a completely smooth build. A few things slowed me down or made me rethink my approach, and I think they're worth documenting honestly rather than glossing over:

- Designing dashboards that actually communicate: choosing which visuals to use, picking a colour palette that read clearly, and laying charts out so the dashboard told a story instead of turning into a wall of numbers took more iteration than I expected — it's easy to add charts, harder to make them work together.
- Keeping KPI values consistent across three tools: with the same figures appearing in Excel, SQL, and Power BI, I had to cross-check every headline number between them to make sure I wasn't presenting three different "truths" for the same metric.
- Chasing down small rounding differences: a few totals were off by a small amount between tools, and it took some digging to confirm this was purely a decimal-to-whole-number display difference between Excel, SQL Server, and Power BI, not a mistake in my calculations (see the Validation Note in Section 2).
- Learning several tools at once: Pivot Tables and Pivot Charts, Power BI visuals and DAX measures, and SQL aggregate functions were all things I was applying hands-on for the first time in a project this size, so part of the challenge was simply building the skill while building the report.
- Refining the dashboard through multiple passes: my first dashboard layout wasn't the one that shipped — I went through several rounds of adjusting chart placement, sizing, and colour choices to improve readability and make the KPIs easier to scan at a glance.
- Recovering from a laptop crash mid-project: I lost dashboard work partway through development and had to rebuild it from scratch. Once recreated, I went back and validated every KPI and chart again against my SQL and Excel outputs to make sure nothing had changed or been lost in the rebuild.

10. Key Learnings

Beyond the finished report, this project taught me a lot that I couldn't have picked up from tutorials alone:

- Built interactive dashboards from scratch in both Excel and Power BI, including KPI cards, slicers, and multiple chart types working together on one page.
- Strengthened my SQL by writing real aggregate queries — SUM, AVG, COUNT, MIN/MAX — along with GROUP BY, filtering, and TOP-N ranking logic across sales, product, region, customer, payment, and time dimensions.
- Learned to validate results across tools rather than trusting a single output — cross-checking Excel, SQL, and Power BI against each other caught the rounding issue early and gave me more confidence in every number I reported.
- Developed business storytelling skills: moving from a raw number to "why this matters" and then to a specific recommendation is a different skill from just running a query, and this project is where I actually practised it end-to-end.
- Improved my dashboard design instincts — layout, colour choice, and avoiding clutter — through direct trial and error rather than following a template.
- Strengthened my overall analytical thinking by working through a full project lifecycle: cleaning data, exploring it, validating it, visualising it, and finally explaining it in business terms.

11. Limitations

As with any analysis, the scope and conclusions in this report are bounded by the data and files available:

- Two-year window: the dataset covers FY 2024–2025 only, which is enough for a like-for-like year-over-year comparison (Section 7.8) but too short to distinguish a genuine multi-year trend from normal year-to-year variation.
- Single denormalised table: all analysis is based on one flat transaction table with no separate customer, product, or store dimension tables — sufficient for the category, region, and time-based analysis in this report, but not for relationship-based modelling (e.g. Power BI star-schema relationships) or customer-level tracking.
- No unique customer identifier: the data distinguishes only New vs. Regular customer type, not individual customers, so repeat-purchase rate and customer lifetime value could not be calculated.
- Power BI reviewed via screenshot: without the underlying .pbix file, the specific DAX formula text behind each KPI card could not be confirmed; the Power BI section (Section 6) describes the data model and visuals based on what is directly observable and cross-validated against Excel/SQL totals.
- Descriptive and diagnostic scope only: this report explains what happened and where, using historical data; it does not include forecasting or predictive modelling.

12. Future Scope

The following extensions would deepen this analysis if additional data or tooling became available:

- Investigate the 2024→2025 softening at a finer grain (by category, state, or store) to identify whether it is concentrated in specific segments or spread evenly across the business.
- Add customer-level (rather than order-level) analysis, such as repeat-purchase rate and customer lifetime value, if a unique customer ID is captured in future.
- Layer in a lightweight forecasting model for the Oct–Nov peak to support more precise inventory and staffing planning.
- Extend the SQL analysis with window functions (e.g. RANK(), ROW_NUMBER()) for deeper category- and region-level ranking without separate queries per group.
- Incorporate additional years of data as they become available, to build a longer, more reliable trend line than the current two-year window allows.
- Rebuild the Power BI data model with proper dimension tables (customer, product, store, date) once available, enabling richer DAX measures and relationship-based analysis.

13. Conclusion

Objectives Achieved

I set out to consolidate 5,000 D-Mart retail transactions into a clean, analysis-ready dataset and analyse it consistently across three tools — Excel (Pivot Tables and Dashboard), SQL (40 business questions across seven analysis groups), and Power BI (interactive dashboard). Every headline figure was cross-validated across all three, and I added a dedicated year-over-year comparison to quantify the change between 2024 and 2025.

Major Findings

The business operates at a healthy ~22% profit margin on ₹63.96 lakh in sales. Grocery leads on volume, Fashion leads on margin, and Maharashtra — Mumbai in particular — carries a disproportionate share of both sales and profit. Regular customers and UPI payments drive most of the business, and October–November stands out as a clear seasonal peak. Year-over-year, 2025 came in modestly softer than 2024 across every core metric — sales, profit, orders, and quantity all dipped slightly, with profit and orders falling a bit faster than sales.

Business Impact

These findings give retail leadership a clear, evidence-based basis for inventory planning, regional investment, payment-experience priorities, and customer-retention strategy, while the year-over-year comparison flags an early, measurable signal worth investigating before it compounds into a larger trend. Throughout, the analysis replaces intuition with a consistent, three-tool view of the same underlying data.

Closing Note

I built this report to hold up under scrutiny — every figure ties back to a specific pivot table, SQL query, or dashboard visual, and I've tried to be upfront about the challenges and limitations in Sections 9 and 11 rather than glossing over them. The recommendations in Section 8 and the extensions in Section 12 (Future Scope) outline where I'd take this next if I kept building on it. Overall, this project pushed me to work through a complete analytics workflow end-to-end, and it's the piece of work I'm most confident walking through in an interview.